

PQC in Digital Certificates: The Transition to ML-DSA and SLH-DSA in X.509

Day 26 | Category: Web Browser and PKI Security

Date: 2026-09-19 |

DAILY MONITORING

PQC in Digital Certificates: The Transition to ML-DSA and SLH-DSA in X.509

The transition to Post-Quantum Cryptography (PQC) fundamentally impacts the Public Key Infrastructure (PKI) that underpins internet security. Digital certificates (X.509), which validate the identity of websites, servers, and applications, have historically relied on classical algorithms like RSA and ECDSA for digital signatures. As these are vulnerable to quantum attacks, the migration to quantum-resistant signature schemes is a critical step in securing future communications.

Key Insight: Why Digital Certificates are Crucial

Digital certificates are the bedrock of trust online. They bind a public key to an identity, verified by a trusted Certificate Authority (CA). If the signature algorithm used to validate this binding is broken, an attacker could forge certificates, impersonate legitimate services, and intercept encrypted traffic. This makes the transition to PQC signatures a non-negotiable requirement for long-term internet security.

The New Standards: ML-DSA and SLH-DSA

NIST has standardized two primary PQC signature algorithms to replace their classical counterparts:

- **ML-DSA (Module-Lattice-based Digital Signature Algorithm):** Defined in FIPS 204, ML-DSA is expected to be the primary replacement for ECDSA due to its efficiency and relatively small signature sizes. It is well-suited for general-purpose use where performance is a key consideration.
- **SLH-DSA (Stateless Hash-based Digital Signature Algorithm):** Defined in FIPS 205, SLH-DSA offers a high level of security based on the well-understood properties of hash functions. Its primary drawback is its larger signature size and slower performance compared to ML-DSA. It is often considered for high-assurance applications like firmware signing or root CA certificates where security assurance is paramount.

Comparing PQC and Classical Signature Schemes

The move to PQC signatures involves significant trade-offs in performance and size. The table below compares ML-DSA and SLH-DSA with the widely used classical algorithm ECDSA (P-256).

Metric	ECDSA (P-256)	ML-DSA-65 (Level 3)	SLH-DSA-128s (Level 1)
Security Basis	Elliptic Curve Discrete Logarithm	Module Learning with Errors (MLWE)	Hash Functions (SHA-256)
Public Key Size	~64 bytes	~1.3 kB	~32 bytes
Signature Size	~72 bytes	~2.9 kB	~7.8 kB
Performance	Very Fast	Fast (slower than ECDSA)	Slow
Quantum Resistance	No	Yes	Yes

Internet Applicability: Hybrid Certificates and Crypto-Agility

The transition is not a flip-the-switch event. Initially, we are seeing the deployment of **hybrid certificates**. These certificates contain signatures from both a classical algorithm (like ECDSA) and a PQC algorithm (like ML-DSA). This ensures backward compatibility with older systems while providing forward security against quantum threats. Certificate Authorities (CAs) and browser vendors are actively testing and deploying these hybrid models to ensure a smooth transition without breaking existing infrastructure.

Daily Quiz

1. What is the primary role of a digital signature in an X.509 certificate?
2. Why is ML-DSA generally preferred over SLH-DSA for common applications like web server certificates?
3. What is a hybrid certificate and why is it important for the PQC migration?